

Bluestock Mutual Fund Analytics

Capstone Project Report

Amisha

July 2026

Table of Contents

- 1. Executive Summary
- 2. Introduction & Objectives
- 3. Data & Methodology
- 4. Key Findings
 - 4.1 AUM Trends
 - 4.2 NAV Performance
 - 4.3 SIP Inflows
 - 4.4 Scheme Performance
 - 4.5 Investor Behavior
- 5. Visualizations & Tables
- 6. Risk & Limitations
- 7. Recommendations
- 8. Appendix

1. Executive Summary

The Indian mutual fund industry, represented across the 40 schemes analyzed in this project, has nearly doubled in assets under management between 2022 and 2025, driven by sustained retail participation through systematic investment plans. Industry-wide AUM grew **107.9%**, rising from ₹30.18 lakh crore to ₹62.74 lakh crore over the period, while monthly SIP inflows grew **169.2%**, reaching ₹31,002 crore by the most recent month with 9.35 crore active SIP accounts. Investor folios more than doubled, from 13.26 crore to 26.12 crore. Small Cap funds delivered the strongest 1-year returns among the 40 schemes studied but also carried the highest tail risk, while Liquid funds delivered the most consistent risk-adjusted performance. A notable concern surfaced in investor behavior data: the vast majority of long-term SIP investors show irregular contribution patterns rather than disciplined monthly investing.

Key findings:

- **AUM growth of 107.9%** industry-wide (₹30.18L Cr → ₹62.74L Cr, Mar 2022–Dec 2025), with SBI Mutual Fund leading at ₹12.5L Cr in AUM, followed by ICICI Prudential (₹10.74L Cr) and HDFC (₹9.3L Cr).
- **Top 1-year performers are concentrated in Small Cap:** ABSL Small Cap Fund (+24.93%) and SBI Small Cap Fund (+24.56%) lead all 40 schemes, but Small Cap funds also register the deepest historical VaR (95%) at -2.69% daily, confirming a clear risk-return trade-off.
- **SIP inflows grew 169.2%**, from ₹11,517 Cr (Jan 2022) to ₹31,002 Cr (latest month), with active SIP accounts reaching 9.35 crore — evidence of sustained retail participation even through market volatility.

- **Liquid funds dominated FY25 category inflows** at ₹4,51,275 Cr net, well ahead of Sectoral/Thematic (₹1,03,829 Cr) and Flexi Cap (₹63,989 Cr), suggesting investors parked significant short-term capital in low-risk instruments during the year.
- **SIP continuity is a material risk:** among investors with 6+ SIP transactions, **97.8%** show an average gap exceeding 35 days between installments (mean $\approx$ 65 days) — far from the ~30-day cadence a disciplined monthly SIP implies, pointing to a broad retention/engagement gap rather than an isolated issue.

2. Introduction & Objectives

Audience: This report is prepared for a senior investment committee reviewing the health, performance, and investor engagement trends of the Indian mutual fund industry, using a representative sample of 40 schemes across Equity and Debt categories.

Objectives:

1. Quantify industry-wide growth in AUM, SIP inflows, and investor folios over 2022–2025.
2. Evaluate scheme-level performance (returns, risk, Sharpe ratio) against benchmarks.
3. Profile investor behavior — cohort patterns, SIP discipline, and continuity risk.
4. Surface actionable, prioritized recommendations for product strategy and investor engagement.

Scope: Analysis draws on 10 processed datasets covering fund master data, daily NAV history (2022–2026), AMC-level AUM, monthly SIP inflows, category-level inflows, industry folio counts, scheme performance metrics, investor transaction records (32,779 transactions across 5,000 investors), portfolio holdings, and benchmark index levels.

3. Data & Methodology

Data sources (10 processed CSV files):

File	Contents	Rows
01_fund_master_cleaned.csv	Scheme identifiers, category, plan, fund manager, risk grade	40
02_nav_history_cleaned.csv	Daily NAV per scheme, 2022–2026	46,000
03_aum_by_fund_house_cleaned.csv	Quarterly AUM by AMC	91
04_monthly_sip_inflows_cleaned.csv	Industry-wide monthly SIP inflow & account counts	49
05_category_inflows_cleaned.csv	Monthly net inflow by category	145
06_industry_folio_count_cleaned.csv	Industry folio counts by segment	22
07_scheme_performance_cleaned.csv	Returns, Sharpe, Sortino, std dev, ratings per scheme	40

08_investor_transactions_cleaned.csv	Investor-level SIP/Lumpsum/Redemption transactions	32,779
09_portfolio_holdings_cleaned.csv	Stock/sector-level portfolio weights per scheme	323
10_benchmark_indices_cleaned.csv	Daily benchmark index levels (NIFTY50, NIFTY100, etc.)	8,051

Cleaning & transformations:

- Daily returns derived from NAV history via percentage change per scheme (`amfi_code` - partitioned).
- AUM and SIP figures aggregated to industry level by summing across AMCs/months.
- Category inflows filtered to India’s FY convention (April–March) to compute FY25 totals.
- Investor cohorts assigned by year of first recorded transaction per `investor_id`.
- Risk metrics (VaR, CVaR, HHI) computed directly from raw NAV and holdings data rather than taken from pre-aggregated fields, to ensure methodology transparency.

4. Key Findings

4.1 AUM Trends

Headline: Industry AUM nearly doubled in under four years, led by the largest incumbent AMCs.

Total AUM across all fund houses grew from ₹30.18 lakh crore (Mar 2022) to ₹62.74 lakh crore (Dec 2025) — a **107.9%** increase. SBI Mutual Fund holds the largest share at ₹12.5 lakh crore, followed by ICICI Prudential (₹10.74L Cr), HDFC (₹9.3L Cr), Nippon India (₹7.0L Cr), and Kotak Mahindra (₹5.8L Cr). This concentration among the top 5 AMCs suggests scale and brand trust remain dominant factors in investor fund-house selection.

Table 1 — AMC AUM Ranking (latest quarter)

Fund House	AUM (₹ Cr)
SBI Mutual Fund	12,50,000
ICICI Prudential MF	10,74,000
HDFC Mutual Fund	9,30,000
Nippon India MF	7,00,000
Kotak Mahindra MF	5,80,000

Recommended visual: Line chart (AUM trend) + horizontal bar chart (AMC ranking) — see `chart_aum_trend.png`.

4.2 NAV Performance

Headline: Small Cap funds led 1-year returns, but with proportionally higher volatility and tail risk.

ABSL Small Cap Fund (+24.93%) and SBI Small Cap Fund (+24.56%) posted the highest 1-year returns among all 40 schemes. However, the same Small Cap category occupies 6 of the 7 riskiest positions by historical VaR (95%), with SBI Small Cap Fund (Direct) showing the worst daily VaR at -2.69%. At the other end, Liquid funds (e.g., Kotak Liquid Fund at +4.26% 1-year return) delivered the lowest returns but also the lowest risk, and ICICI Pru Liquid Fund posted the highest Sharpe ratio in the entire dataset (7.68) — reflecting a low-volatility, consistent-return profile rather than genuinely superior performance.

Recommended visual: Line chart, NAV (indexed to 100) vs NIFTY 100 benchmark — see [chart_nav_vs_benchmark.png](#).

4.3 SIP Inflows

Headline: SIP inflows grew 169% over the period, reflecting deepening retail participation.

Monthly SIP inflows rose from ₹11,517 crore (Jan 2022) to ₹31,002 crore in the latest month, a **169.2%** increase, with active SIP accounts reaching 9.35 crore. At the category level, Liquid funds attracted the largest FY25 net inflow (₹4,51,275 Cr), followed by Sectoral/Thematic (₹1,03,829 Cr) and Flexi Cap (₹63,989 Cr) — indicating investors used liquid funds heavily for short-term parking of capital during FY25 even as long-term SIP participation grew.

Recommended visual: Bar chart, monthly SIP inflow — see [chart_sip_inflows.png](#).

4.4 Scheme Performance

Headline: Risk-grade classifications are internally consistent with realized Sharpe ratios, validating their use in recommendation logic.

Across the 40 schemes, Liquid funds cluster at the “Low” risk grade with Sharpe ratios up to 7.68 and near-zero annualized volatility; Large Cap funds cluster at “Moderate” with Sharpe ratios around 1.0; and Mid/Small Cap funds cluster at “High”/“Very High” with Sharpe ratios in the 0.8–0.96 range but materially higher volatility (19–25% annualized std dev). Axis Bluechip Fund carries the highest expense ratio in the dataset at 1.64%.

Recommended visual: Scatter plot, Return (X) vs Std Dev (Y), bubble size = AUM, colored by category.

4.5 Investor Behavior

Headline: SIP continuity is weak across the investor base — a retention risk, not an isolated pattern.

Among the 1,362 investors with 6 or more SIP transactions, **97.8%** show an average gap between installments exceeding 35 days (mean gap $\approx$ 65 days), well beyond the $\sim$ 30-day cadence implied by disciplined monthly investing. New investor cohorts (2024: 4,803 investors; 2025: 197 investors) both gravitate toward higher-growth categories — HDFC Mid-Cap Opportunities and SBI Small Cap Fund respectively — as their most-preferred fund, rather than more conservative options.

Recommended visual: Histogram of SIP gap-days distribution, with a reference line at 30 days.

5. Visualizations & Tables

#	Chart	Source File	Takeaway
1	Industry AUM Trend (line)	03_aum_by_fund_house_cleaned.csv	AUM nearly doubled (+107.9%) in under 4 years
2	NAV vs Benchmark (line, indexed)	02_nav_history_cleaned.csv , 10_benchmark_indices_cleaned.csv	Actively managed fund tracked/outperformed NIFTY 100 over the period
3	Monthly SIP Inflows (bar)	04_monthly_sip_inflows_cleaned.csv	SIP inflows grew 169% driven by rising active account base
4	Return vs Risk scatter	07_scheme_performance_cleaned.csv	Small Cap = highest return & risk; Liquid = highest Sharpe, lowest risk
5	SIP Gap-Days Histogram	08_investor_transactions_cleaned.csv	97.8% of long-term SIP investors exceed the 35-day continuity threshold

6. Risk & Limitations

- **Sample scope:** Analysis covers 40 schemes and 5,000 investors — a representative subset, not the full AMFI universe; absolute AUM/inflow figures should be read as directional, not exhaustive industry totals.
- **Transaction data window:** Investor transaction records span Jan 2024–May 2025 only, limiting cohort analysis to two first-transaction-year groups (2024, 2025); a longer window would allow multi-year cohort comparison.
- **VaR/CVaR are historical, not forward-looking:** both metrics assume the past return distribution is representative of future risk, which may understate risk during unprecedented market conditions.
- **HHI concentration** is computed from a single portfolio snapshot per fund (latest available date) rather than a time series, so it reflects current, not historical, concentration.
- **SIP continuity threshold (35 days)** is a reasonable but somewhat arbitrary cutoff; sensitivity analysis at different thresholds (e.g., 40, 45 days) would strengthen the at-risk classification.

7. Recommendations

- 1. **(High priority) Launch a SIP re-engagement program** targeting the 97.8% of long-term SIP investors showing irregular contribution gaps — automated reminders or incentives tied to consecutive on-time installments could materially improve retention.
- 2. **(High priority) Segment new-investor onboarding by risk appetite**, since both 2024 and 2025 cohorts default toward aggressive growth funds — early risk-profiling could reduce mismatch between investor risk tolerance and fund choice.
- 3. **(Medium priority) Review expense ratios on high-cost, moderate-Sharpe funds** such as Axis Bluechip Fund (1.64% expense ratio) relative to peers with comparable risk-adjusted returns at lower cost.
- 4. **(Medium priority) Flag equity funds with high sector concentration** (e.g., Axis Bluechip Fund, HHI $\approx$ 2,968) for diversification review, since concentration risk does not correlate cleanly with category label.
- 5. **(Lower priority) Extend the investor transaction data window** in future analysis cycles to enable multi-year cohort tracking and more robust continuity-rate trending.

8. Appendix

8.1 Data Dictionary (key fields)

Field	Description
amfi_code	Unique scheme identifier (AMFI code)
nav	Net Asset Value per unit, daily
var_95_daily_pct	5th percentile of daily return distribution (Historical VaR)
cvar_95_daily_pct	Mean of returns at/below the VaR threshold (Expected Shortfall)
sharpe_ratio	Risk-adjusted return metric (annualized)
hhi_index_0_10000	Herfindahl-Hirschman sector concentration index
avg_gap_days	Average days between consecutive SIP transactions per investor
at_risk	Flag: TRUE if avg_gap_days > 35

8.2 Sample SQL Query — Top 5 Schemes by 1-Year Return

```
SELECT category
FROM
ORDER BY
LIMIT 5
```

8.3 Sample SQL Query — Monthly SIP Inflow Trend

```
SELECT month
FROM
ORDER BY month ASC
```

8.4 File References

- Report generated from: 01_fund_master_cleaned.csv through 10_benchmark_indices_cleaned.csv
- Supporting analytics notebook: Advanced_Analytics.ipynb
- Supporting output files: var_cvar_report.csv , sector_hhi.csv , sip_continuity.csv , cohort_analysis.csv